

Commerce XI Maths App

Activities 1.1–1.4

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 20

Question 1 - MCQ · PG-21–23 · ACT-1.1 · Step 1

To divide ordered marks into four equal parts, use

Options

- A. quartiles
- B. deciles
- C. percentiles
- D. the mean

Correct answer: quartiles

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives quartiles.

Question 2 - MCQ · PG-21–23 · ACT-1.1 · Step 2

The three values required to divide marks into four equal parts are

Options

- A. Q_1 , Q_2 and Q_3
- B. D_1 , D_2 and D_3
- C. P_1 , P_2 and P_3
- D. mean, median and mode

Correct answer: Q_1 , Q_2 and Q_3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_1 , Q_2 and Q_3 .

Question 3 - MCQ · PG-21–23 · ACT-1.1 · Step 3

To divide ordered marks into ten equal parts, use

Options

- A. deciles
- B. quartiles
- C. percentiles
- D. class marks

Correct answer: deciles

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives deciles.

Question 4 - MCQ · PG-21–23 · ACT-1.1 · Step 4

The number of decile values used to divide data into ten equal parts is

Options

- A. 9
- B. 10
- C. 4
- D. 99

Correct answer: 9

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 9.

Question 5 - MCQ · PG-21–23 · ACT-1.1 · Step 5

Before finding partition values for classroom marks, first arrange the marks in

Options

- A. ascending order
- B. random order
- C. alphabetical order
- D. descending frequency

Correct answer: ascending order

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives ascending order.

Question 6 - MCQ · PG-21–23 · ACT-1.2 · Step 1

For marks 18, 24, 31, 35, 42, the number of students is

Options

- A. 5
- B. 4
- C. 6
- D. 150

Correct answer: 5

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 5.

Question 7 - MCQ · PG-21–23 · ACT-1.2 · Step 2

For marks 18, 24, 31, 35, 42, the average mark is

Options

- A. 30
- B. 24
- C. 31
- D. 35

Correct answer: 30

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 30.

Question 8 - MCQ · PG-21–23 · ACT-1.2 · Step 3

For marks 18, 24, 31, 35, 42, the median mark is

Options

- A. 31
- B. 24
- C. 30
- D. 35

Correct answer: 31

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 31.

Question 9 - MCQ · PG-21–23 · ACT-1.2 · Step 4

For marks 18, 24, 31, 35, 42, the student with the middle score obtained

Options

- A. 31 marks
- B. 24 marks
- C. 30 marks
- D. 35 marks

Correct answer: 31 marks

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 31 marks.

Question 10 - MCQ · PG-21–23 · ACT-1.2 · Step 5

The score below which 75% of ordered observations lie is

Options

A. Q_3

B. Q_1

C. Q_2

D. D_3

Correct answer: Q_3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_3 .

Question 11 - MCQ · PG-21–23 · ACT-1.3 · Step 1

Students with attendance below 50% are placed in the

Options

A. black list

B. quartile list

C. merit list

D. frequency list

Correct answer: black list

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives black list.

Question 12 - MCQ · PG-21–23 · ACT-1.3 · Step 2

The partition value corresponding to 50% is

Options

A. Q_2 or P_{50}

B. Q_1 or P_{25}

C. Q_3 or P_{75}

D. D_1 or P_{10}

Correct answer: Q_2 or P_{50}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_2 or P_{50} .

Question 13 - MCQ · PG-21–23 · ACT-1.3 · Step 3

To estimate the number of students below 50% attendance graphically, use an

Options

- A. ogive
- B. pie chart only
- C. pictograph
- D. simple bar chart only

Correct answer: ogive

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives ogive.

Question 14 - MCQ · PG-21–23 · ACT-1.3 · Step 4

On a less than ogive, the number below 50% attendance is read on the

Options

- A. Y-axis
- B. X-axis
- C. title
- D. class-width column

Correct answer: Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Y-axis.

Question 15 - MCQ · PG-21–23 · ACT-1.3 · Step 5

On the ogive, begin at attendance 50 on the X-axis and draw a line parallel to the

Options

- A. Y-axis
- B. X-axis
- C. class interval
- D. curve

Correct answer: Y-axis

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Y-axis.

Question 16 - MCQ · PG-21–23 · ACT-1.4 · Step 1

Six equally placed fire exits divide the 114-floor building into

Options

- A. 7 approximately equal sections
- B. 6 sections
- C. 10 sections
- D. 100 sections

Correct answer: 7 approximately equal sections

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 7 approximately equal sections.

Question 17 - MCQ · PG-21–23 · ACT-1.4 · Step 2

For 114 floors divided by six internal partition points, the first ideal position is approximately

Options

- A. 16.3
- B. 19
- C. 28.5
- D. 57

Correct answer: 16.3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 16.3.

Question 18 - MCQ · PG-21–23 · ACT-1.4 · Step 3

A practical first fire-exit floor nearest to $114/7$ is

Options

- A. 16
- B. 7
- C. 19
- D. 57

Correct answer: 16

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 16.

Question 19 - MCQ · PG-21–23 · ACT-1.4 · Step 4

Using multiples of approximately $114/7$, a suitable set of six exit floors is

Options

- A. 16, 33, 49, 65, 81, 98
- B. 19, 38, 57, 76, 95, 114
- C. 10, 20, 30, 40, 50, 60
- D. 25, 50, 75, 100, 110, 114

Correct answer: 16, 33, 49, 65, 81, 98

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 16, 33, 49, 65, 81, 98.

Question 20 - MCQ · PG-21–23 · ACT-1.4 · Step 5

The mathematical idea used to place the fire exits is

Options

- A. partitioning an interval into equal parts
- B. finding only the mean
- C. random sampling
- D. drawing a pie chart

Correct answer: partitioning an interval into equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives partitioning an interval into equal parts.